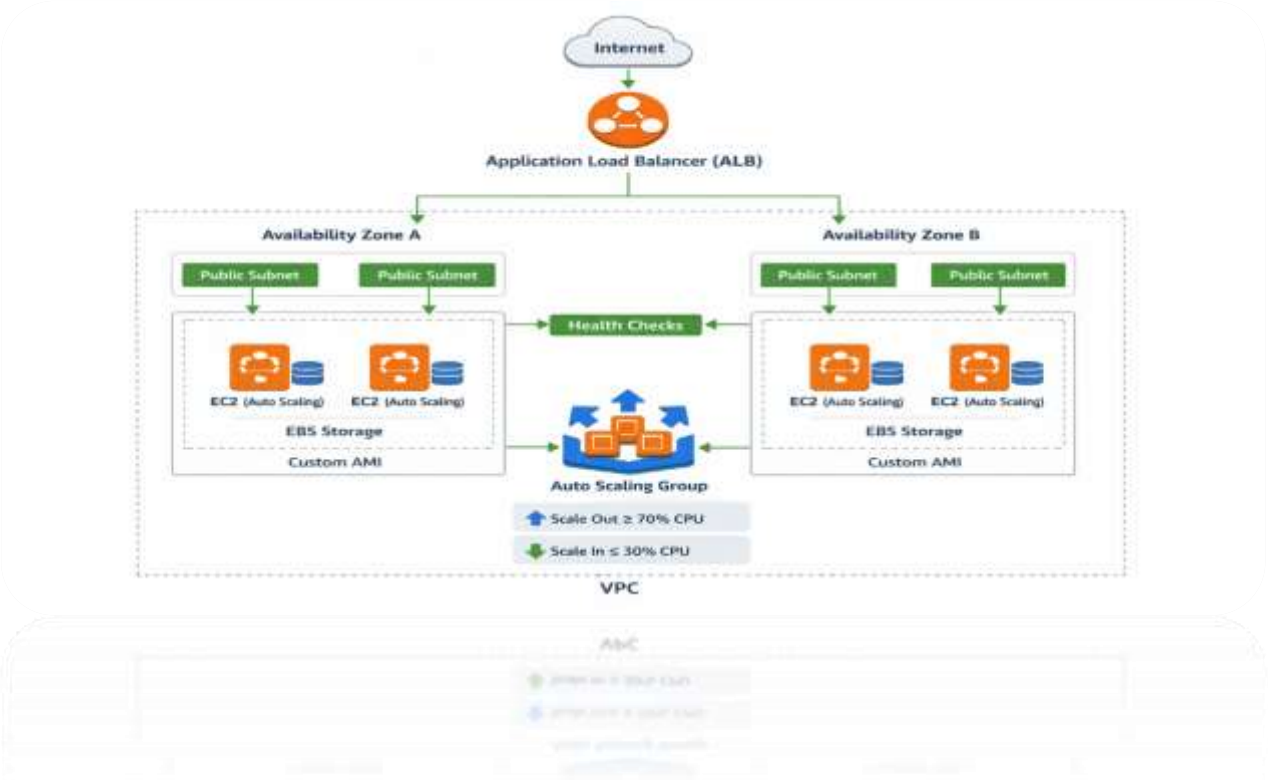
Highly Available Web Application with Auto Scaling

STEP 1: Create VPC (Multi-AZ Architecture)



1. Go to **VPC Dashboard**
2. Click **Create VPC**
   - Name: Task1-VPC
   - CIDR: 10.0.0.0/16
   - Enable DNS Hostnames

Create Subnets (Minimum 2 AZs)

Subnet	AZ	CIDR	Type
Public-Subnet-A	ap-south-2a	10.0.1.0/24	Public
Public-Subnet-B	ap-south-2b	10.0.2.0/24	Public

Attach Internet Gateway

- Create IGW
- Attach to VPC
- Update Public Route Table:
    - Destination: 0.0.0.0/0 - IGW

Now VPC spans 2 AZs - High Availability achieved.

STEP 2: Launch Base EC2 & Install Web Server

1. Go to EC2 - Launch Instance
2. Select:
    - Amazon Linux 2
    - t2.micro
    - Public subnet
    - Enable Auto-assign Public IP
    - 10GB EBS
3. Configure Security Group:
    - Allow HTTP (80) from Anywhere
    - Allow SSH (22) from Your IP
4. SSH into instance and install Apache:

```
#!/bin/bash

sudo apt -y update
sudo apt -y install apache2
sudo systemctl start apache2
sudo systemctl enable apache2
sudo apt -y install git
sudo rm -rf /var/www/html/*
sudo mkdir -p /var/www/html/
sudo git clone   link  /var/www/html/
```

5. Verify using Public IP

STEP 4: Create Launch Template

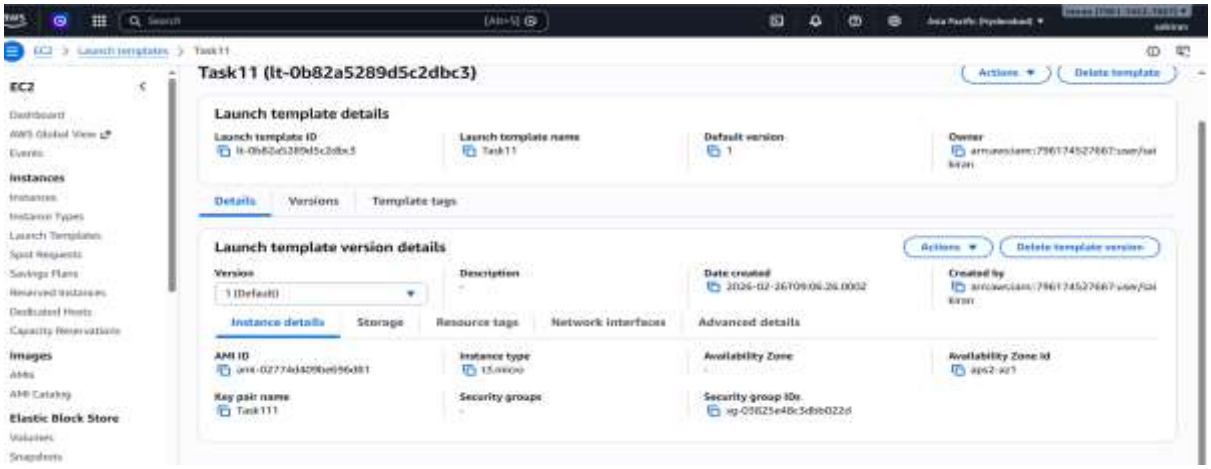
Go to EC2 → Launch Templates → Create

Configure:

- AMI - Custom AMI created
- Instance Type - t2.micro
- Key pair - Select
- Security Group - WebApp-SG
- Storage - 8GB EBS
- Versioning - Default v1

Click Create.

Launch Template created successfully.



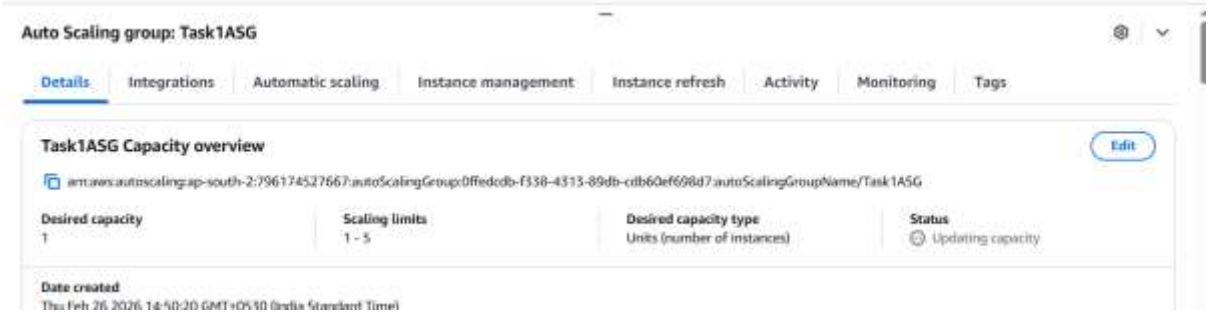
STEP 6: Create Target Group

1. EC2 - Target Groups - Create
2. Target type - Instances
3. Protocol - HTTP
4. Port - 80
5. VPC - Task1-VPC

Health Check Settings:

- Protocol - HTTP
- Path - /
- Interval - 30 seconds
- Timeout - 5 seconds
- Healthy threshold - 3
- Unhealthy threshold - 2

Create Target Group.

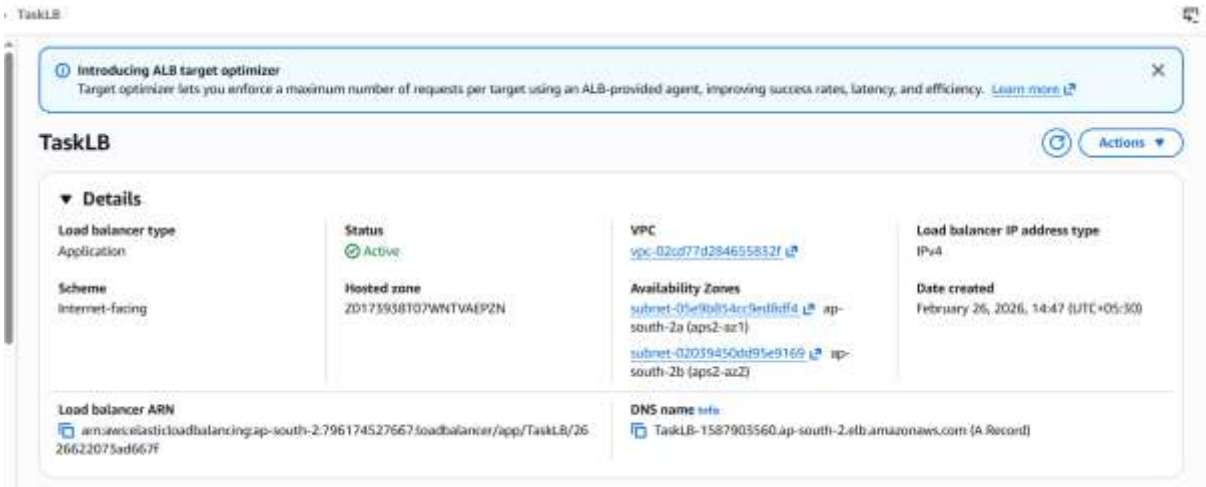


STEP 7: Create Application Load Balancer

1. EC2 - Load Balancers → Create
2. Type - Application Load Balancer
3. Scheme - Internet-facing
4. Select both Public Subnets (AZ A & AZ B)
5. Attach ALB-SG
6. Listener - HTTP (80)
7. Forward to Target Group

Click Create.

ALB deployed across 2 AZs.



STEP 8: Create Auto Scaling Group (ASG)

1. Go to EC2 → Auto Scaling Groups
2. Click Create
3. Choose Launch Template
4. Select VPC
5. Select Private Subnet A & B

6. Attach to Target Group

Capacity Configuration:

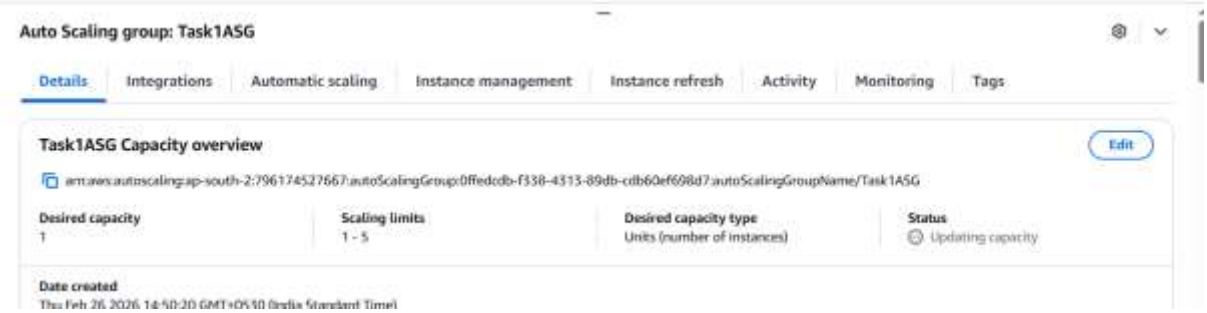
- Minimum: 2
- Desired: 2
- Maximum: 5

Health Check:

- Type - ELB + EC2
- Grace Period - 300 seconds

Create ASG.

Now 2 instances automatically launch in 2 AZs.



STEP 9: Configure Scaling Policies

Go to ASG → Automatic Scaling

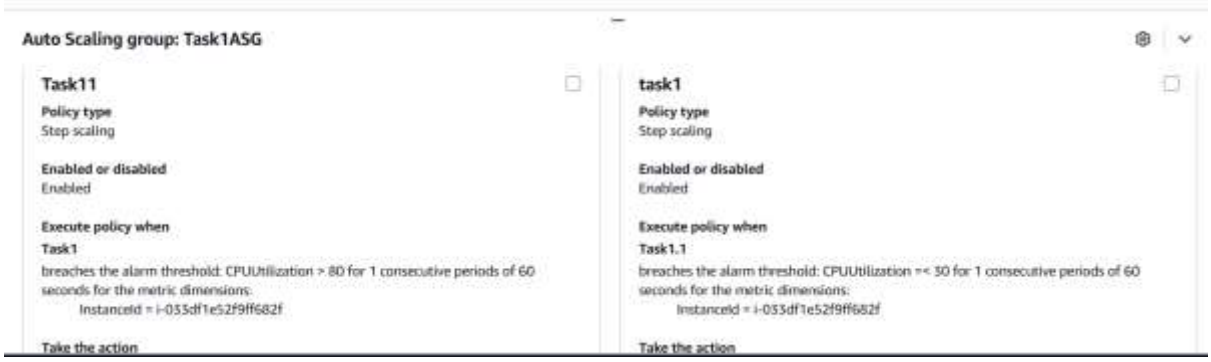
Scale Out Policy

- Type - Target Tracking or Step Scaling
- Metric - CPU Utilization
- Threshold - 70%
- Action - Add 1 instance

Scale In Policy

- Threshold - 30%
- Action - Remove 1 instance

Metric source: CloudWatch



Why ALB Not NLB?

In this project, we used an **Application Load Balancer (ALB)** instead of a **Network Load Balancer (NLB)** because the application is a web-based HTTP application.

Layer Difference

- **Application Load Balancer** - Works at **Layer 7 (Application Layer)**
- **Network Load Balancer** - Works at **Layer 4 (Transport Layer)**

Since our application uses **HTTP**, we need Layer 7 features.